SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0515** **Slot B**

COURSE OUTCOME COVERED
CO4: ***Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*** **Assessment Tool Weightage: 40%**

QUESTIONS:5 **Total Marks 50** **Duration :60 Minutes**

Annexure A
Industry Problem Solving Task

Code of Conduct:

I M. Pranay (192525382)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay

Rubrics for Evaluation

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	***Marks(100)***
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy ≥ 90%
- Latency ≤ 80 ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms
Method B: 91%, 85 ms
Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

Answer:

The traffic monitoring system has three important requirements:

- Accuracy ≥ 90%
- Latency ≤ 80 ms
- Works under varying lighting conditions

Method	**Accuracy**	**Latency**	**Evaluation**
Method A	88%	60 ms	Fails accuracy requirement
Method B	91%	85 ms	Fails latency requirement
Method C	92%	75 ms	Satisfies both

Analysis:

- **Method A** has acceptable latency of 60 ms, but its accuracy is only 88%, which is below the required 90%.
- **Method B** provides 91% accuracy, but its latency is 85 ms, exceeding the maximum limit of 80 ms.
- **Method C** provides 92% accuracy and 75 ms latency, satisfying both requirements.

Conclusion:

Method C is the best choice because it satisfies all specified constraints. It provides accuracy above 90% and latency below 80 ms, making it the most suitable method for real-time traffic surveillance. Its suitability under varying lighting should also be validated using representative traffic data.

2. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy ≥ 85%

Options:

Model A: 83%, 40 ms
Model B: 87%, 65 ms
Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

Answer:

The drone vision system requires:

- Lightweight computation
- Real-time detection below 60 ms
- Accuracy ≥ 85%

Model	**Accuracy**	**Detection Time**	**Evaluation**
Model A	83%	40 ms	Fails accuracy

Model	Accuracy	Detection Time	Evaluation
Model B	87%	65 ms	Fails real-time requirement
Model C	86%	55 ms	Satisfies both

Analysis:

- **Model A** is fast at 40 ms, but its accuracy is only 83%, which is below the required 85%.
- **Model B** provides 87% accuracy but takes 65 ms, exceeding the 60 ms real-time limit.
- **Model C** provides 86% accuracy and takes only 55 ms, satisfying both accuracy and real-time requirements.

Conclusion:

Model C is the most suitable choice for the drone vision system. It achieves the required accuracy while maintaining detection time below 60 ms. It is therefore a better balance between computational efficiency, accuracy, and real-time operation.

3. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

Answer:

The security system requires:

- High accuracy greater than 92%
- Moderate computational requirements

Approach	Accuracy	Computation	Evaluation
Traditional	85%	Low	Accuracy too low
Deep Learning	94%	High	High accuracy but high computation
Hybrid	91%	Moderate	Accuracy below required level

Analysis:

- **Traditional methods** require low computation, but their accuracy of 85% is insufficient for a high-security application.
- **Deep Learning** achieves 94% accuracy, which exceeds the required 92%. However, it requires high computational resources.
- **Hybrid methods** require moderate computation, but their accuracy of 91% does not meet the required accuracy.

Conclusion:

Among the given options, **Deep Learning is the best approach** because it is the only method that achieves the required accuracy of more than 92%. Although its computational requirement is high, accuracy is critical in a security system. If moderate computation is an absolute constraint, none of the given options completely satisfies both requirements, so optimization or a more efficient deep-learning model would be needed.

4. Motion Tracking in Sports Analytics
Requirements:

- Accurate motion tracking
- Real-time performance

Options:
Optical Flow: accurate, slow
Motion Estimation: fast, less accurate
Compare and evaluate the suitable approach. Justify using constraints.
Answer:
The sports analytics system requires:

- Accurate motion tracking
- Real-time performance

The two available approaches are:

Technique	Accuracy	Speed	Evaluation
Optical Flow	Accurate	Slow	High accuracy but unsuitable for strict real-time use
Motion Estimation	Less accurate	Fast	Suitable for real-time but may reduce accuracy

Analysis:
Optical Flow provides accurate motion tracking, making it useful when tracking precision is the highest priority. However, its slow processing makes it difficult to satisfy real-time requirements.
Motion Estimation provides faster processing and is therefore better suited for real-time sports applications. However, its lower accuracy can affect the quality of motion analysis.
Conclusion:
For a system where **real-time performance is essential**, **Motion Estimation is the more suitable choice**. However, its accuracy should be improved through model optimization, filtering, or combining it with more accurate tracking techniques. If accuracy is more important than real-time speed, Optical Flow would be preferable.

5. Face Recognition System Design
Constraints:

- Accuracy ≥ 90%
- Works in low light
- Moderate computation

Answer:
The face recognition system has the following constraints:

- Accuracy ≥ 90%
- Works in low-light conditions
- Moderate computation

Approach	Accuracy	Evaluation
Traditional	82%	Does not satisfy accuracy requirement
Deep Learning	93%	Satisfies accuracy requirement
Hybrid	89%	Slightly below required accuracy

Analysis:

- **Traditional methods** provide only 82% accuracy, which is significantly below the required 90%.
- **Deep Learning** provides 93% accuracy, exceeding the required 90%, and can be designed to handle variations such as low-light images.
- **Hybrid methods** provide 89% accuracy, which is slightly below the required threshold.

The main disadvantage of Deep Learning is its potentially higher computational requirement. For moderate computation, a lightweight deep-learning architecture or optimized model can be considered.
Conclusion:
Deep Learning is the best choice among the given options because it achieves 93% accuracy, which is above the

required 90%. For low-light operation, the system should be trained or enhanced using low-light face images. A lightweight or optimized deep-learning model can help satisfy the moderate-computation requirement.

Options:
Traditional: 82%
Deep Learning: 93%
Hybrid: 89%
Compare and evaluate the best choice. Justify logically.